



Investigating the Day-Ahead Electricity Market: Transformer-Based Forecasting of Sharp Price Movements and Day-Ahead Prices

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August 2026

Abstract

This report presents the work completed during a research internship at PHAETHON Centre of Excellence. The project investigated the use of transformer-based deep-learning models to analyse and forecast sharp movements in Day-Ahead electricity prices. The work progressed from controlled experiments on synthetic time series to applications using real electricity-market data. Its main contributions were a binary transformer classifier for predicting Jump and No Jump market intervals and a Cyprus price forecaster that used predicted Jump events to segment the target day and forecast prices within each segment.

Project Aim and Objectives

The main aim of this project was to investigate whether transformer-based models can identify and forecast abrupt changes in Day-Ahead electricity prices, with particular emphasis on the timing of sharp movements in the Cyprus electricity market.

The original research proposal approached this question through the study of process-like behaviour in electricity-price time series. The completed project further developed this idea by moving from broad stochastic-process classification to the more targeted prediction of individual Jump and No Jump intervals. These predictions were subsequently used to structure the price-forecasting task.

The specific objectives were:

- to develop an understanding of transformer architectures for time-series classification and forecasting;
- to test transformer models using controlled synthetic processes with known labels;
- to adapt the resulting methods to real electricity-market data;

- to investigate whether market forecast variables contain information about subsequent price Jumps;
- to develop a binary classifier that predicts Jump and No Jump intervals;
- to use predicted jump probabilities to identify meaningful boundaries within a forecast day; and
- to develop and evaluate a segmented transformer-based price forecaster for the Cyprus Day-Ahead Market.

Background and Motivation

The liberalisation of electricity markets began in the 1990s with the aim of increasing competition, improving market efficiency, and promoting the participation of new market actors. This transition marked a shift from vertically integrated monopoly structures to competitive market environments, creating new opportunities for electricity producers, suppliers, traders, and consumers. While market liberalisation has enhanced efficiency, it has also introduced greater levels of uncertainty and risk.

Unlike most commodities, electricity cannot be economically stored on a large scale, requiring generation and consumption to remain continuously balanced in real time. Any imbalance between supply and demand can lead to significant fluctuations in electricity prices.

Electricity markets are consequently characterised by high price volatility, making electricity one of the most volatile traded commodities. Electricity price behaviour is influenced by a wide range of factors, including fuel prices, weather conditions, electricity demand, generation availability, transmission constraints, carbon-emission allowance prices, and the growing penetration of renewable energy sources (RES).

The Day-Ahead Electricity Market

The Day-Ahead Market is a market in which electricity is traded in advance, shortly before it is physically delivered. Market participants buy and sell electricity for specified delivery periods on the following day.

The Cyprus electricity market has operated within a liberalised framework since October 2025 (Konstantinou, 2025). Its Day-Ahead Market is organised as a two-sided auction, in which buyers and generators submit demand and supply orders specifying quantities and, where applicable, the prices at which they are willing to trade. These are matched to determine the Market Clearing Price for each of the 48 half-hourly periods of the following delivery day (*The New Electricity Market Arrangements in Cyprus*, 2015).

The forecasting problem concerns not only the general daily price level but also the timing of price movements within the day. Accurately identifying the timing of price movements may be relevant for generation planning, bidding strategies, exposure management, and short-term risk assessment.

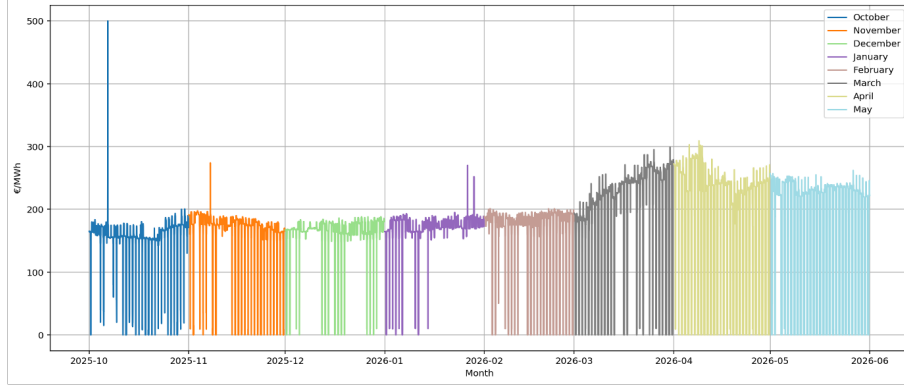


Figure 1: Cyprus Day-Ahead electricity prices over the period October 2025–May 2026. The series displays substantial variation over time, characterised by frequent sharp downward movements and occasional price spikes.

Figure 1 illustrates that the Cyprus Market Clearing Price does not evolve smoothly. The series contains sudden declines, sharp recoveries, and occasional extreme observations. These characteristics suggest that electricity prices cannot be adequately represented by one simple, smooth stochastic process, and motivate the use of flexible time-series models capable of learning complex behaviour directly from data.

Cyprus Market Price Behaviour and Research Motivation

Exploratory analysis of the Cyprus market reveals a recurring daily pattern. Prices commonly decline during the morning or middle of the day as photovoltaic generation increases, before rising again later in the afternoon as photovoltaic output falls and other forms of generation are required to meet demand. Figure 2 illustrates this relationship using the average behaviour of the Market Clearing Price and the main electricity-market quantities, and provides the principal empirical motivation for the project.

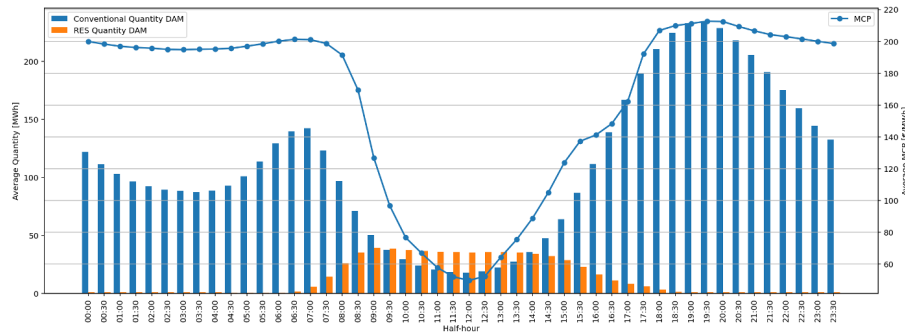


Figure 2: Average daily behaviour of the Cyprus Market Clearing Price and electricity-market quantities. The average price declines during periods of increased photovoltaic production and rises as photovoltaic output falls later in the day.

The figure shows a pronounced average daily price shape, but this average conceals substantial day-to-day variation. On individual days, transitions may occur earlier or later and may be more or less pronounced. Some days also contain other abrupt movements or extreme spikes. Consequently, reproducing the average daily curve is not sufficient: a forecast may approximate the overall daily price level and shape while placing the main decline or increase in the wrong

half-hour. The forecasting objective therefore extends to predicting the timing and severity of these movements on a particular day, which is particularly relevant because electricity-market decisions apply to specific half-hourly delivery periods.

Transformer-Based Time-Series Modelling

The methodological focus of the project was the use of transformer-based deep-learning models for time-series analysis. Transformers were introduced by Vaswani et al. (2017) and have since attracted considerable research interest due to their ability to learn complex relationships within sequential data. Although they were originally developed for natural-language processing, their ability to model relationships across a sequence also makes them relevant for time-series classification and forecasting.

The central mechanism of a transformer is attention. Attention allows the model to compare observations at different positions in an input sequence and learn which of them are most relevant to the task. In electricity-market applications, this may help the model identify short-term dependence, repeated daily patterns, earlier price transitions, or relationships between market forecast variables.

A transformer uses positional encoding to preserve the order of the observations. This is necessary because the attention mechanism does not otherwise distinguish automatically between values observed recently and those observed further in the past.

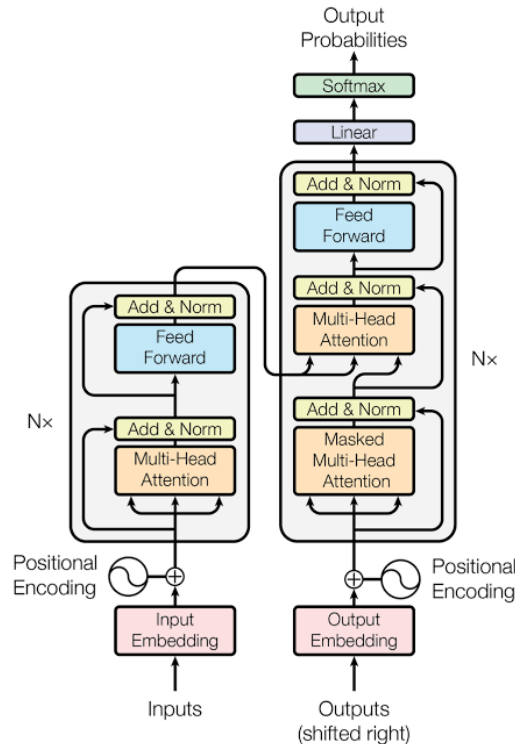


Figure 3: Standard transformer architecture introduced by Vaswani et al. (2017) showing the encoder and decoder components.

The encoder analyses the input sequence and constructs an internal representation of its relevant relationships. A decoder may then use this representation, together with previously known or

generated output values, to produce a future sequence. During autoregressive generation, each predicted value is fed back to the decoder as an input for predicting the next value.

An encoder-only transformer can be appropriate when an input sequence is mapped to a class label or when the entire future horizon is predicted directly in one operation. An encoder-decoder transformer can be appropriate when the model generates a future sequence using the encoded historical information and earlier output values.

During training, model predictions are compared with observed targets using a loss function and model parameters are adjusted to reduce the prediction error.

Project Description

The project developed a transformer-based framework for analysing and forecasting electricity-market time series. Its central application was the Cyprus Day-Ahead Market, where prices are determined for 48 half-hourly delivery periods. German electricity-market data was also used as an additional case study to examine whether parts of the methodology transferred to a larger hourly market. Cyprus was the principal case study and received the most extensive evaluation.

The work followed a staged learning approach organised into seven chronological stages. Stages 1-3 established the transformer methodology using synthetic data, where the underlying behaviour and class labels were known. Stage 1 introduced transformer-based time-series forecasting through a noisy sine-wave example. Stage 2 developed an encoder-only transformer that classified five simulated stochastic-process families. Stage 3 combined process classification with multi-step sequence forecasting.

Stages 4 and 5 transferred these ideas to real electricity-market data. Cyprus and German price series were analysed, market-oriented process classifiers were trained, and price-forecasting models were developed. Additional experiments examined transition-aware forecasting and the inclusion of richer market information, including Cyprus forward-contract quantity.

Stages 6 and 7 formed the main research contribution. Stage 6 reformulated sharp-movement prediction as an interval-level binary classification problem. For every half-hourly Cyprus interval or hourly German interval, an encoder-only transformer predicted a jump probability from market forecast variables, such as expected load and renewable generation. A selected cut-off value, called the jump-probability threshold, converted this probability into a final Jump or No Jump label. The original labels were derived from changes in realised prices, whereas the transformer inputs consisted of forecast market variables rather than the realised price itself. A separate classifier was trained for each target date using only observations from before that date, while the complete target day was excluded from training and used for testing. However, some preprocessing and historical-probability construction steps used information that was not organised in a fully chronological manner.

Stage 7 connected Jump classification with price forecasting. The Cyprus jump probabilities produced in Stage 6 were used to identify possible changes during each forecast day. The initial approach divided the day at individual predicted Jump boundaries, and a separate encoder-only transformer produced the complete price forecast for each resulting section. Because treating every predicted Jump as a separate boundary produced excessive segmentation, a subsequent

collapse-aware approach was developed. This method grouped neighbouring high-probability intervals into broader event windows and selected either the beginning and end of each window or its single highest-probability half-hour as candidate segment boundaries.

Preliminary Development: Stages 1-5

The first five stages established the necessary transformer, classification, forecasting, and real-data processing methods. Stages 6 and 7 then developed the main Jump-classification and segmented-forecasting frameworks. Because the final two stages represent the central contribution of the project, Stages 1-5 are summarised briefly below and Stages 6 and 7 are discussed in greater detail in the following sections.

Stage 1: Transformer Forecasting Fundamentals

Stage 1 introduced transformer-based time-series forecasting using a generated noisy sine wave. The development of the models was informed by an online article and its accompanying source-code repository (Michaels, 2023; S., 2024). Three models of increasing complexity were implemented: an encoder-only model for next-value prediction, an encoder-decoder model for single-value prediction, and an encoder-decoder model for multi-step sequence forecasting.

This stage established the use of time-series windows, positional encoding, attention, training procedures, and forecast evaluation. It served as a methodological foundation and did not provide a direct data or model input to the later stages.

Stage 2: Stochastic-Process Classification

Stage 2 examined whether an encoder-only transformer could distinguish between five simulated forms of time-series behaviour: Geometric Brownian Motion, Ornstein-Uhlenbeck mean reversion, jump behaviour, GARCH-like volatility clustering, and seasonal mean reversion. Since each sequence was generated computationally, its true class was known and classification performance could be assessed reliably in a controlled setting.

The encoder processed each input window, meaning a fixed-length, consecutive section of a longer time series. It then produced an internal representation of the behaviour contained in that window. A classification layer converted this representation into one unrestricted numerical score, known as a logit, for each process class. Softmax converted these logits into class probabilities that summed to one. The class with the highest probability became the predicted process label.

This stage introduced transformer-based sequence classification and established the idea of identifying process-like behaviour from an observed time-series window.

Stage 3: Joint Classification and Forecasting

Stage 3 combined classification and forecasting in a process-aware encoder-decoder transformer. The model received a historical sequence, classified its process-like behaviour, and forecasted a

sequence of future values.

The encoder analysed the historical observations, while the decoder used the encoded representation and shifted target values to generate the future sequence. During training, target shifting taught the decoder to predict each future value from the correct preceding value. A separate classification head used only the encoder representation to predict the simulated process family, whereas the forecasting branch used both the encoder and decoder. During evaluation, the decoder was also given shifted ground-truth targets. Consequently, each prediction was conditioned on the correct preceding value rather than on the model’s previous prediction. The reported forecasting performance therefore reflects teacher-forced evaluation rather than fully autoregressive forecasting.

Training combined forecasting and classification losses, encouraging the encoder to learn representations useful for both tasks. This stage therefore examined whether recognising the type of process-like behaviour and forecasting its continuation could be learned jointly.

Stage 4: Initial Application to the Cyprus Market

Stage 4 transferred the earlier methods from controlled simulated data to real half-hourly Cyprus Day-Ahead electricity prices. An initial experiment reused the model trained in Stage 3. The Stage 3 model had been jointly trained on simulated data to classify stochastic-process behaviour and forecast a future sequence. For the Stage 4 experiment, its weights were frozen and only the encoder and classification head were used. This frozen classification component was applied to real Cyprus price windows without retraining to determine which of the five simulated process families each window most closely resembled.

This experiment provided an initial connection between the synthetic and real-market stages, but it also exposed a mismatch between the conditions under which the Stage 3 model had been trained and those under which it was applied. In particular, the model was trained on general textbook simulations rather than simulations designed specifically to resemble half-hourly electricity prices. Consequently, the frozen model provided an exploratory baseline rather than a market-specific classifier.

A new market-oriented encoder-only classifier was trained specifically for the Cyprus application. It retained the same five benchmark classes but used newly generated artificial data that was better aligned with half-hourly electricity prices. For example, to represent the daily structure of the Cyprus market, one simulated process reproduced a repeating daily pattern based on 48 half-hourly observations, with random variation added to prevent each day from being identical. After training on the labelled artificial data, the classifier was applied to real Cyprus price windows.

The resulting classifications continued to represent similarity to simplified simulated benchmarks. For example, a real window classified as “jump-like” contained patterns that the model associated with the simulated jump class. Because real-market data does not contain objectively known process labels, the outputs remain exploratory model-based indications rather than verified market regimes.

Stage 4 also developed a price-only encoder-decoder transformer that used 96 historical half-hourly prices, corresponding to two days, to forecast the following 48 half-hourly prices. Histor-

ical price was the only model input. The model was trained using shifted target prices, while validation and test forecasts were generated autoregressively from the model’s own preceding predictions. Later experiments investigated the timing of the main price transitions and whether explicitly accounting for the morning decline and afternoon increase could represent the daily price curve more accurately.

Stage 5: Market and Feature Extensions

Stage 5 extended the real-market analysis through transition-aware forecasting, the inclusion of additional model inputs, known as features, and application to German electricity data. A feature is an input variable supplied to a model to help it make a prediction. Historical price is one feature, while forecast load, renewable generation, production, consumption, or forward-contract quantity are examples of additional features that may contain information relevant to future prices.

The first Cyprus transition-aware approach defined the main morning decline as the largest negative half-hourly price change between 07:00 and 12:00 and the main afternoon increase as the largest positive change between 12:00 and 19:00. A transformer encoder used the previous seven days, consisting of 336 half-hourly prices, to predict probability distributions over the possible decline and increase times for the following day. The predicted transition times were then used to shift and stretch sections of the previous day’s price curve, aligning its main decline and increase with the expected transition times.

A second experiment used a joint encoder-decoder transformer to generate a sequence of 48 next-day prices. Its encoder processed seven historical days and predicted distributions for the timing of the morning decline and afternoon increase. During training, the decoder received shifted ground-truth prices. During validation and forecasting, it generated prices autoregressively: each predicted price was fed back into the decoder to predict the next half-hourly price. The decoder also used the encoded historical information, predicted timing distributions, and clock information.

A further Cyprus experiment compared the price-only forecaster with a model that also used historical forward-contract energy quantity as an input. The new model used the previous 96 half-hourly prices together with the forward-contract quantity recorded during each of those past intervals to forecast prices for the next 48 half-hours. This investigated whether historical information connected with the Forward market could add value to Day-Ahead price forecasting. However, the available variable represented the quantity of energy associated with the forward contracts rather than a Forward market price, so the experiment provided only a partial test of the broader relationship proposed between the two markets.

The market-oriented classification and price-forecasting workflow was also applied to German hourly electricity data. As in the retrained Cyprus classification experiment, a new encoder-only classifier was trained on artificial, market-oriented time series and then applied to real German price windows. To represent the daily structure of the German market, one simulated process used a repeating 24-period daily cycle. A separate price-only forecaster used 48 historical hourly prices to forecast the following 24 hourly prices. Like the original Cyprus forecaster, it used historical Day-Ahead price as its only predictive feature.

A rich-feature German forecaster was then developed as a comparison with this price-only model. For the historical 48-hour input period, it used past Day-Ahead prices together with forecast and actual recorded information about renewable production, total production, consumption, residual load, and other production. It also included calendar information representing the hour, day of the week, and month. For the future 24-hour forecast period, the decoder used only market forecasts and calendar information that would be known in advance. Recorded production and consumption values for that future period were excluded because they would not have been available when making the forecast. This experiment examined whether the additional market information improved forecasting beyond the information contained in historical prices alone.

Stage 6: Jump Classification

Stage 6 reformulated the analysis of sharp price movements as a binary classification problem. Each market interval was assigned one of two classes:

- No Jump, represented by 0; or
- Jump, represented by 1.

An interval corresponded to one half-hour in the Cyprus data and one hour in the German data. The classifier produced a jump probability for each interval, which was converted into a final Jump or No Jump label using a selected probability threshold.

The central distinction in this stage was between the information used to construct the historical labels and the information supplied to the transformer. Realised electricity prices were used to identify whether a historical interval contained an unusually large price change. However, realised prices were not used as transformer input features. The model instead attempted to predict the price-derived label from forecast variables available before the realised market price.

Construction of Original Jump Labels

The original Jump and No Jump labels were generated from consecutive changes in realised electricity prices. For interval t , the price difference was calculated as:

$$\Delta P_t = P_t - P_{t-1},$$

where P_t is the realised market price at interval t .

The mean $\mu_{\Delta P}$ and standard deviation $\sigma_{\Delta P}$ of the price differences were calculated across the supplied market dataset. An interval was labelled as a Jump when the price difference deviated from the mean by more than k standard deviations:

$$|\Delta P_t - \mu_{\Delta P}| > k\sigma_{\Delta P}.$$

Otherwise, it was labelled No Jump. The multiplier k controlled the strictness of the Jump definition; the implementation used $k = 1.5$. The use of an absolute difference meant that both

large upward and large downward price changes were included in the Jump class. The classifier did not distinguish the direction or magnitude of the movement.

One limitation of the current construction is that the mean and standard deviation used to define the labels were calculated from the complete supplied workbook. A stricter chronological design could recalculate the label boundary using only the history available before each target date.

Input Features

The model inputs consisted of forecast electricity-market conditions rather than realised prices. The five Cyprus features were:

- load forecast;
- photovoltaic generation forecast;
- wind generation forecast;
- total renewable-energy forecast; and
- forecasted residual load.

The five German features represented:

- total renewable-production forecast;
- total production forecast;
- total consumption forecast;
- forecasted residual load; and
- other production forecast.

These variables were hypothesised to provide indirect information about the risk of a future price Jump. For example, high photovoltaic production relative to demand could be associated with downward price movement. Conversely, high demand and limited renewable production could indicate that more expensive generation will be required to meet demand, increasing the possibility of a sharp price rise. Nevertheless, the model could learn only statistical associations between the forecasted conditions and the historical Jump labels. It did not observe the previous price level or price trajectory from which a change would occur.

Because the features were measured on different numerical scales, each feature was standardised by subtracting its training-set mean and dividing by its training-set standard deviation. The same training statistics were then used to standardise the validation and test rows, preventing information from the target day from influencing preprocessing.

Rolling Train, Validation and Test Splits

A rolling evaluation procedure was used to preserve the chronological nature of the forecasting task, meaning that later observations were not used to predict an earlier date. It is called rolling because the target date moved forward one day at a time, with a separate classifier trained and tested for each date. All permitted observations before each target date formed the historical

candidate data, while the complete target day was excluded from training and reserved as an untouched test set.

For Cyprus, the historical sample began on 1 October 2025, while the rolling evaluation covered target dates from 1 April to 31 May 2026. A classifier evaluated on 1 April was trained using observations from 1 October 2025 up to, but not including, 1 April. When the target date moved to 2 April, a new classifier was additionally allowed to use the observations from 1 April as part of its training history. Each target date contained 48 half-hourly test observations.

For Germany, a rolling one-year historical window was used, with 24 hourly observations retained for each target date.

After class balancing, 70% of the historical samples were used for training and 30% for validation. Although all these observations occurred before the test date, the split between training and validation was random rather than strictly chronological. The training set was used to update the model parameters, while the validation set was used to compare training epochs and retain the best-performing version of the model. The complete target day formed the test set and was used only for the final evaluation.

Class Balancing

Jump observations were considerably less common than No Jump observations. A classifier trained on the unbalanced data could therefore obtain apparently high accuracy by predicting No Jump for most intervals.

To reduce this effect, the historical data was balanced separately for each target date. The majority No Jump class was randomly downsampled to the size of the minority Jump class. The resulting balanced collection was then divided into training and validation sets. The target-day test set was not balanced and retained the real class distribution.

Balancing ensured that the classifier encountered equal numbers of Jump and No Jump examples during training. However, it also discarded many valid No Jump observations. Alternatives for future work include class-weighted loss, weighted sampling, or training on all observations followed by threshold optimisation.

Transformer Architecture and Training

The principal Stage 6 method used an encoder-only transformer and was row-wise: one market interval formed one sample and produced one Jump or No Jump prediction. Each sample contained five standardised features and entered the transformer with the shape:

[1 sequence position, 5 features]

A linear input layer projected the five features into a 32-dimensional internal representation. This representation passed through two transformer encoder layers with four attention heads, a 64-dimensional feed-forward component, layer normalisation, and dropout. Mean pooling then averaged the encoded representations across the sequence to produce one overall summary for classification. However, because the row-wise model contained only one sequence position,

this operation summarised a single encoded interval. A final classification layer converted this summary into two logits: one for no jump and one for jump.

Self-attention normally allows each position in a sequence to use information from other positions in that sequence. However, the principal row-wise model received only one interval at a time, giving it a sequence length of one. Its attention layer could therefore attend only to the single position. The model could still learn relationships among the five forecast variables through its input-projection and feed-forward layers, but it could not compare the current interval with earlier or later half-hours or hours. An exploratory daily-sequence model instead received all 48 Cyprus half-hours, or all 24 German hours, together. In this version, self-attention could relate the forecasted conditions at one interval to the pattern across the rest of the day. The daily-sequence approach was retained as an exploratory comparison, while the row-wise method remained the principal Stage 6 workflow and supplied the probabilities used in Stage 7.

The model was trained for ten epochs using cross-entropy loss, the Adam optimiser, a learning rate of 0.001, and a batch size of 64. One epoch represented one complete pass through the training data. After each epoch, performance was measured on the validation set.

For the reported Stage 6 runs, the model state with the highest validation accuracy was retained as a checkpoint, meaning a file containing the trained model parameters and related settings. The implementation used for those runs updated the retained state only when an epoch achieved a strictly higher validation accuracy. Consequently, if multiple epochs achieved the same highest accuracy, the earlier of those epochs was retained. Validation loss was calculated but was not used to resolve ties.

The current implementation retains validation accuracy as the default checkpoint-selection metric but also permits validation Jump F1 to be selected. The model state with the highest value of the selected metric is retained. If two epochs achieve the same selected metric value, the state with the lower validation cross-entropy loss is retained. The optional Jump F1 metric and loss-based tie-break were introduced after the reported Stage 6 runs and therefore did not affect the checkpoints used to produce the reported results.

Probability Threshold and Final Transformer Labels

Softmax converted the two output logits produced by the classification layer into corresponding probabilities. The final label was determined by comparing the jump probability with a threshold τ :

$$\text{Jump if } P(\text{jump}) \geq \tau; \quad \text{No Jump otherwise.}$$

Increasing the threshold made the classifier more conservative, producing fewer Jump predictions. Lowering it detected more real Jumps but also increased the number of false alarms. The principal fixed thresholds were 0.9095 for Cyprus and 0.7035 for Germany. These were project-specific operating choices selected through trial-and-error testing, with the aim of approximately matching the number of transformer-predicted Jump labels to the number of original Jump labels.

The threshold affected only the conversion from probability to label; it did not change the trained model or its probability outputs.

Evaluation Metrics

The classifier was evaluated at different jump-probability thresholds using the predicted and original label counts, accuracy, Jump precision, Jump recall, Jump F1, and confusion matrices.

Accuracy measured the proportion of intervals classified correctly. However, it was not sufficient on its own because No Jump was the dominant class; a model could achieve relatively high accuracy by labelling many No Jumps while failing to detect many original Jumps. Jump precision measured the proportion of predicted Jumps that were correct, while Jump recall measured the proportion of original Jumps successfully detected. The Jump F1 score combined precision and recall into a single measure. The confusion matrices provided further information by distinguishing false Jumps from missed Jumps:

Original label	Predicted label	Interpretation
No Jump	No Jump	Correct No Jump
No Jump	Jump	False Jump
Jump	No Jump	Missed Jump
Jump	Jump	Correct Jump

Table 1: Interpretation of Jump classification outcomes.

These errors were particularly relevant to Stage 7. A false Jump could introduce an unnecessary Stage 7 segment, whereas a missed Jump could remove a potentially useful boundary.

Cyprus Threshold Comparison and Results

The Cyprus threshold comparison covered 2,928 half-hourly observations from April and May 2026. Among these observations, 199 were originally labelled Jump and 2,729 were labelled No Jump.

The following table reports the main comparison thresholds:

Threshold	Accuracy	Jump Precision	Jump Recall	Jump F1	Predicted Jumps	Correct Jumps	False Jumps	Missed Jumps
0.7000	0.7579	0.1860	0.7588	0.2987	812	151	661	48
0.8000	0.8074	0.1902	0.5628	0.2843	589	112	477	87
0.9000	0.8712	0.1523	0.1960	0.1714	256	39	217	160
0.9095	0.8788	0.1139	0.1156	≈ 0.1147	202	23	179	176

Table 2: Jump classification performance for different probability thresholds.

At a threshold of 0.70, the classifier detected 151 of the 199 original Jumps, giving a recall of approximately 76%. However, it also produced 661 false Jumps and substantially overestimated the total number of Jump intervals.

At 0.9095, the predicted Jump count was 202, which was close to the original count of 199. Nevertheless, the apparent agreement was misleading. The classifier correctly detected only 23 Jumps, missed 176, and produced 179 false Jumps. The false and missed predictions almost cancelled each other in the final count. This demonstrated that matching the total number of predicted Jumps did not imply that the correct intervals were identified. The confusion-count curves in Appendix A show how the number of correct, false, and missed Jump predictions

changed across the tested thresholds.

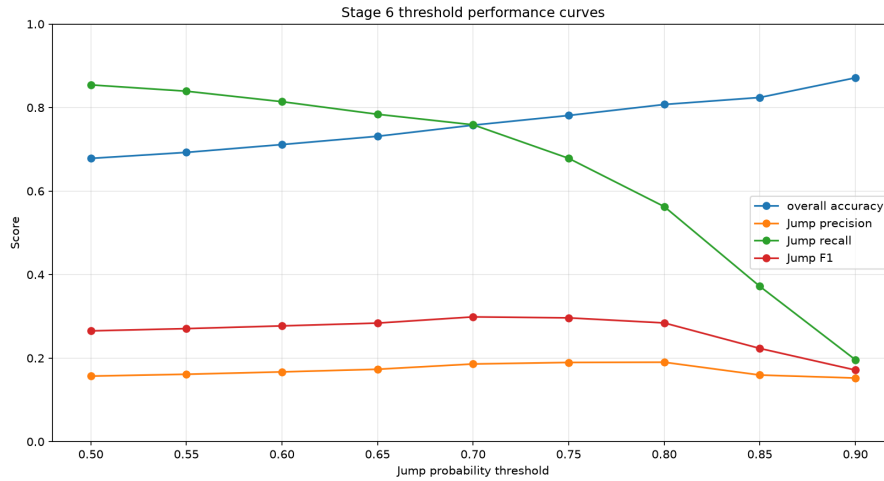


Figure 4: Stage 6 classification performance across different jump-probability thresholds for the Cyprus April–May evaluation period. Increasing the threshold raised overall accuracy but reduced Jump recall, showing that fewer original Jump intervals were detected. Jump precision and F1 remained relatively low across the tested thresholds because many predicted Jumps were false and many original Jumps were missed.

Figure 4 shows the main trade-off created by the probability threshold. Higher thresholds produced greater overall accuracy largely because No Jump was the dominant class. Lower thresholds detected more original Jumps but also produced considerably more false Jumps.

The threshold therefore had to be considered in relation to the Stage 7 forecasting objective. A lower threshold might retain more potentially useful transition information but create excessive segmentation, while a higher threshold could reduce the number of segments at the cost of missing important price movements.

Discussion and Limitations

Stage 6 demonstrated that forecast variables showed some apparent association with subsequent sharp price changes. However, the Cyprus results also revealed a difficult precision-recall trade-off. No tested threshold simultaneously produced high Jump precision and high Jump recall.

The main limitations of Stage 6 were:

- The principal row-wise classifier processed one interval at a time, so its transformer attention mechanism could not compare neighbouring intervals.
- The model did not receive historical prices, recent price changes, explicit time-of-day information, or calendar variables.
- Jump labels were defined using price-difference statistics calculated from the complete workbook rather than only information available before each rolling target date.
- The historical training and validation rows were split randomly after class balancing rather than divided chronologically.
- Balancing produced an artificial 50/50 class distribution and discarded many No Jump observations. Consequently, the softmax scores were not necessarily calibrated to the natural Jump frequency. The softmax outputs should be treated as relative model scores

rather than calibrated real-world probabilities.

- Missing forecast-feature values were interpolated before the rolling splits were constructed. Although only two Cyprus rows were affected, linear interpolation may have used a later observation, so the preprocessing was not strictly limited to information that would have been available at that time.

A threshold-sensitivity analysis was conducted using the saved Cyprus jump probabilities. This analysis did not retrain the transformer. Instead, a range of probability thresholds was applied to the same model outputs, and the resulting accuracy, Jump precision, Jump recall, F1 score, and confusion counts were recalculated. This showed how the model’s final predictions changed as the threshold became more or less strict.

A future model could use complete daily forecast sequences together with leakage-safe historical prices, recent price differences, and calendar variables.

Stage 7: Jump-Segmented Price Forecasting

Stage 7 connected the Stage 6 Jump classifier to the price-forecasting model. For every target day, Stage 6 produced one jump probability for each of the 48 half-hourly periods. Stage 7 used these probabilities to decide where to divide the day before forecasting the Market Clearing Price. The aim was to condition the forecasting model on the predicted timing of Jumps.

The main idea was that different parts of an electricity-market day can behave differently. For example, the overnight period, morning or midday price collapse, afternoon recovery and the period in between may not follow the same price pattern. Instead of forecasting the complete day with one model, Stage 7 divided the day into segments and forecasted each segment separately.

The procedure was:

1. Obtain the 48 Stage 6 jump probabilities for a target day.
2. Use those probabilities to select segment boundaries.
3. Divide the day into price segments.
4. Train one price transformer for each segment.
5. Forecast all prices within each segment.
6. Combine the segment forecasts into one 48-value daily forecast.

The principal evaluation covered the same 61 target dates as the Cyprus Stage 6 rolling experiment, from 1 April to 31 May 2026.

Unlike Stage 6, Stage 7 used historical realised prices directly when training. For each target date, only prices strictly before that date were available to the model. The actual target-day prices were used only after forecasting to calculate evaluation metrics.

Two segmentation approaches were evaluated. The initial approach treated individual Stage 6 Jump predictions as boundaries. The later collapse-aware approach grouped neighbouring high-jump-probability half-hours into event windows before selecting boundaries. Both approaches used the same segment-transformer structure.

Inputs and Segmentation Logic

Stage 7 received historical prices together with load, photovoltaic generation, wind generation, total renewable-energy generation, and residual load forecasts, and Stage 6 jump probabilities for the 14 days preceding the day being forecasted. For the target day, the actual Market Clearing Prices were withheld from the model.

Stage 7 was tested on the dates from 1 April to 31 May 2026. Although the actual prices for these dates were available in the dataset, each date was treated as if it had not happened yet. For each target date, Stage 7 used the saved Stage 6 jump probabilities produced specifically for that date by a classifier trained only on earlier observations. This ensured that each target day received probabilities from a classifier trained only on the history that preceded it.

Segmenting the day was motivated by the presence of distinct daily periods. A single full-day model may smooth the transitions between these periods. Segment-specific models were intended to focus separately on the behaviour within each part of the day.

Segment Transformer and Training Examples

The initial and collapse-aware approaches used the same encoder-only segment-transformer structure. They differed mainly in how the segment boundaries were selected and in the additional forecasting adjustments used by the collapse-aware method, such as analogue blending and clipping, which are explained later. The following training procedure applied to both approaches.

For every target date and segment, Stage 7 constructed training examples from historical days preceding the target date. This historical input period is referred to as the *lookback*, and the standard lookback was 14 days.

For one historical training example, the model received:

- the realised prices, market forecast variables, and the Stage 6 jump probabilities for the same clock segment during the preceding 14 days;
- the market forecast variables and Stage 6 jump probabilities for the segment on the day being predicted; and
- the realised segment prices on the day being predicted, which served as the training target.

For example, the first training example used 1–14 October 2025 to predict 15 October 2025. For a segment covering 10:00–12:00, the model received the realised prices, forecast features, and Stage 6 jump probabilities for 10:00–12:00 on 1–14 October. It also received the forecast features and Stage 6 probabilities for 10:00–12:00 on 15 October. The realised prices for that segment on 15 October were then provided as the training target.

The 15 October prices were included because this was a historical example used to train the model. The model first produced predicted prices, compared them with the known realised prices, and used the resulting error to update its parameters. In contrast, when forecasting the actual Stage 7 target day, such as 1 April 2026, that target day’s realised prices were not provided to the model. They were used only after forecasting to calculate the mean absolute error (MAE) and the root mean squared error (RMSE).

One limitation concerned the Stage 6 jump probabilities used for the historical period from October 2025 to March 2026. Saved Stage 6 jump probabilities were not available for these dates. Stage 7 therefore passed their market data through the Stage 6 model trained for the final target date, 31 May 2026, and used the resulting jump probabilities as price-model inputs. Because that Stage 6 model had learned from some observations occurring after the earlier historical dates, the procedure gave the Stage 7 training data information that would not have been available at the time. This could be corrected by calculating each historical date’s jump probabilities with a Stage 6 classifier trained only on data from before that date.

Each historical day’s segment was flattened into one numerical representation, called a transformer token. The encoder therefore attended across the 14 historical days rather than directly across every half-hour within those days; it compared the historical days with one another. Positional encoding indicated which days were older and which were more recent. Mean pooling then averaged the 14 encoded historical-day tokens to produce one overall historical-context representation.

This historical-context representation was combined with the target-day features for every half-hour in the segment. A direct output head then predicted all prices in the segment simultaneously, without a transformer decoder. It did not predict prices one at a time and did not feed an earlier predicted price back into the model.

A new transformer was trained for every segment and target date. Training used Smooth L1 loss, the AdamW optimiser, a learning rate of 0.0005, a batch size of 16, and 60 epochs. Smooth L1 reduced the influence of unusually large individual errors relative to ordinary squared error.

After each segment was forecasted independently, Stage 7 inserted its predicted values into the corresponding half-hour positions of a 48-value daily array. The completed array formed the target-day forecast. Because each segment was forecasted by a separate transformer, the final prediction of one segment and the first prediction of the next were not required to connect smoothly. This allowed the forecast to represent a sharp price movement at a boundary, which was intentional because each boundary identified a period in which a transition was considered likely. However, independent forecasting could also create an artificial discontinuity if the difference between neighbouring segment predictions did not reflect the actual market behaviour.

A boundary only determined where the day was divided; it did not force the price to change at that exact half-hour. The transformer determined the predicted price pattern within each segment. Therefore, a predicted decline or recovery could occur before or after the selected boundary.

The segment models did not use a separate validation set or validation-based early stopping. The model obtained after the final epoch was therefore always used for forecasting.

Initial Hard-Boundary Approach

The initial approach converted Stage 6 jump probabilities into hard Jump or No Jump labels. Every half-hour classified as Jump became a candidate segment boundary. Boundaries that would create segments shorter than one hour were filtered out, and the beginning and end of the day were added automatically.

For example, retained boundaries at 08:30 and 16:30 divided the target day into three segments: 00:00–08:00, 08:30–16:00, and 16:30–23:30. A separate transformer was trained for each resulting segment.

This method established the complete Jump-segmented forecasting pipeline, but it had two practical problems. First, lower classification thresholds generated many boundaries and therefore required a large number of separate segment-specific transformer training jobs. This substantially increased the runtime. Second, several neighbouring high-probability half-hours could each become a boundary, causing oversegmentation. As a result, one extended price change could be divided into several unnecessarily short segments.

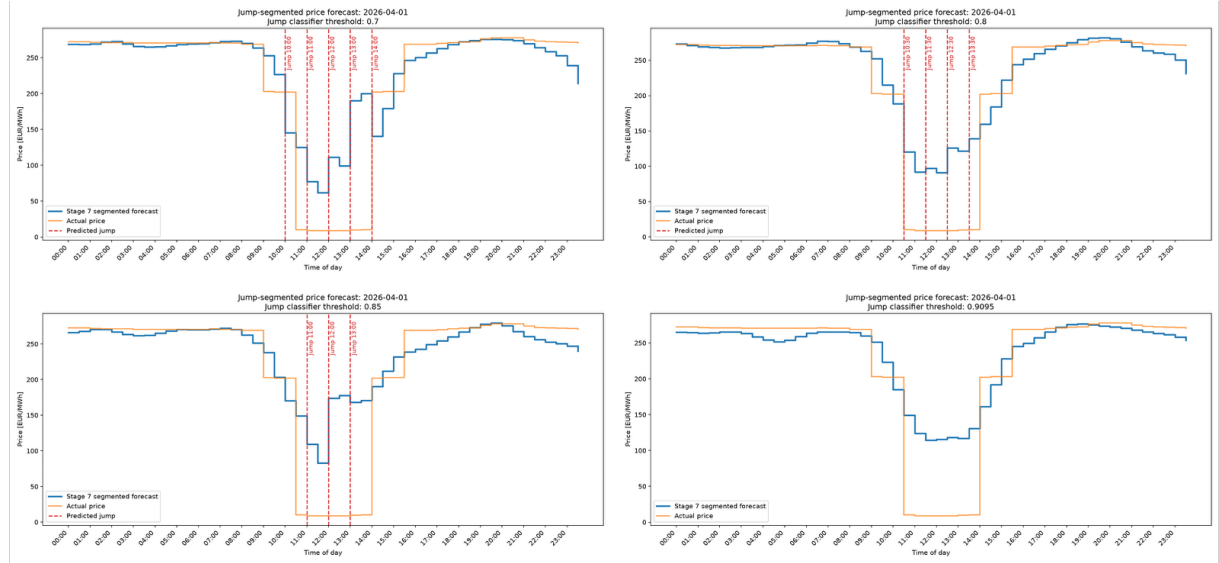


Figure 5: Example of the initial hard-boundary Stage 7 forecast for 1 April 2026, using jump-probability thresholds of 0.70, 0.80, 0.85, and 0.9095. The vertical lines show the half-hourly periods retained as segment boundaries. The forecast illustrates how the original method could require several separately trained segment models and could produce negative predicted prices because historical-range clipping was not applied.

Initial Threshold Comparison

The preliminary comparison of hard segmentation thresholds was limited to the first ten days of April because running every threshold over the complete evaluation period would require substantial computation time.

Threshold	Days	Mean MAE [€/MWh]	Median MAE [€/MWh]	Mean RMSE [€/MWh]	Median RMSE [€/MWh]	Avg predicted Jumps/day	Avg segments/day
0.7	10	26.545	27.248	42.849	41.045	7.500	8.500
0.8	10	25.556	25.668	39.530	39.590	6.500	7.500
0.85	10	25.712	25.661	40.838	37.276	5.300	6.300
0.9095	10	28.277	26.629	43.233	44.692	2.800	3.800

Table 3: Initial threshold comparison over the first ten days of April 2026.

Threshold 0.80 produced the lowest mean MAE and mean RMSE in this preliminary screen. It also required fewer segment models than threshold 0.70. The 0.9095 threshold created substantially fewer segments but had the largest mean forecast errors among the four settings.

This comparison covered only ten days and was therefore treated as an initial diagnostic rather than a final threshold-selection experiment. It showed the relationship between the probability threshold, number of boundaries, runtime, and forecast error.

Collapse-Aware Revision

The collapse-aware method was introduced because neighbouring high-jump-probability half-hours often represented one extended price event rather than several independent events. Instead of applying the Stage 6 hard-label jump-probability threshold to each row independently, the revised approach used a separate event threshold to identify broader high-probability regions. The hard threshold of 0.9095 remained recorded in the outputs because it was the selected Stage 6 classification threshold, but it did not determine the collapse-aware boundaries. The collapse-aware method instead worked directly with the Stage 6 jump probabilities.

All consecutive half-hours whose Stage 6 jump probability reached the event threshold were grouped into one event window. For example, if the probabilities from 10:00 to 12:00 were all at least 0.80, they formed one event rather than four or five separate candidate boundaries.

The distinction between the two thresholds was therefore:

- The Stage 6 hard threshold converted classifier jump probabilities into final Jump or No Jump labels.
- The Stage 7 event threshold identified broader high-probability regions for collapse-aware segmentation.

The start-stop strategy treated the beginning and end of each event window as candidate boundaries. When both boundaries were retained, the complete high-probability period formed one forecast segment.

Candidate boundaries were retained only when they were at least four half-hours, or equivalently two hours, from the day edges and from every previously retained boundary. This minimum spacing could be adjusted through a command-line argument. By default, the implementation retained no more than four event boundaries per day, although this limit could also be changed through a command-line argument. When more than four valid boundaries were identified, the implementation retained the earliest four chronologically rather than selecting the four with the highest Stage 6 jump probabilities. Consequently, an event window could contribute two boundaries, one boundary or no boundary, depending on its position, its distance from the other accepted boundaries, and whether the maximum number of boundaries had already been reached. This logic reduced the risk that one extended price event would be fragmented into several very small forecasting segments.

The default two-hour spacing rule enforced minimum segment length and reduced the number of models that had to be trained. However, it could also discard candidate boundaries associated with short or closely spaced events. For example, the half-hours associated with a brief price spike could receive high Stage 6 jump probabilities without producing a separate forecast segment. This was an important limitation of the default segmentation settings, although the minimum spacing could be reduced in future experiments.

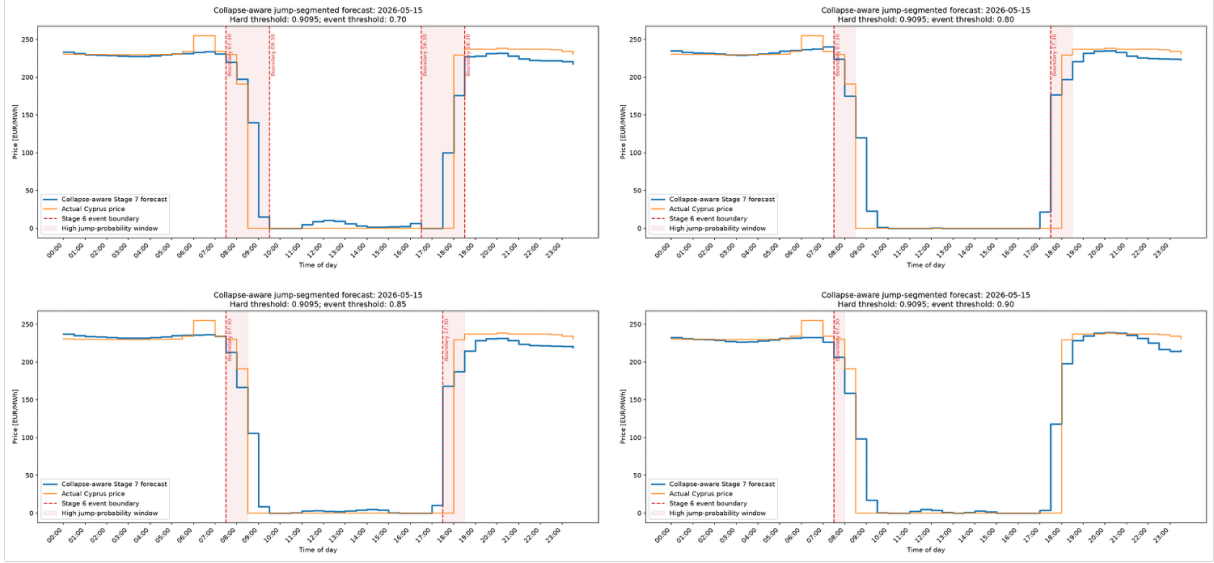


Figure 6: Collapse-aware Stage 7 forecast for 15 May 2026 using start-stop event boundaries of 0.70, 0.80, 0.85, and 0.90. The shaded areas identify high-probability event windows, while the vertical boundary lines show where the retained boundaries were used to divide the target day into independently forecasted segments.

A peak-boundary alternative was also tested. Instead of using the beginning and end of an event window, it considered only the half-hour with the highest Stage 6 jump probability inside that window. This generally reduced the number of candidate boundaries but did not mark both sides of the complete event window. The candidate was still subject to the minimum-spacing and maximum-boundary rules.

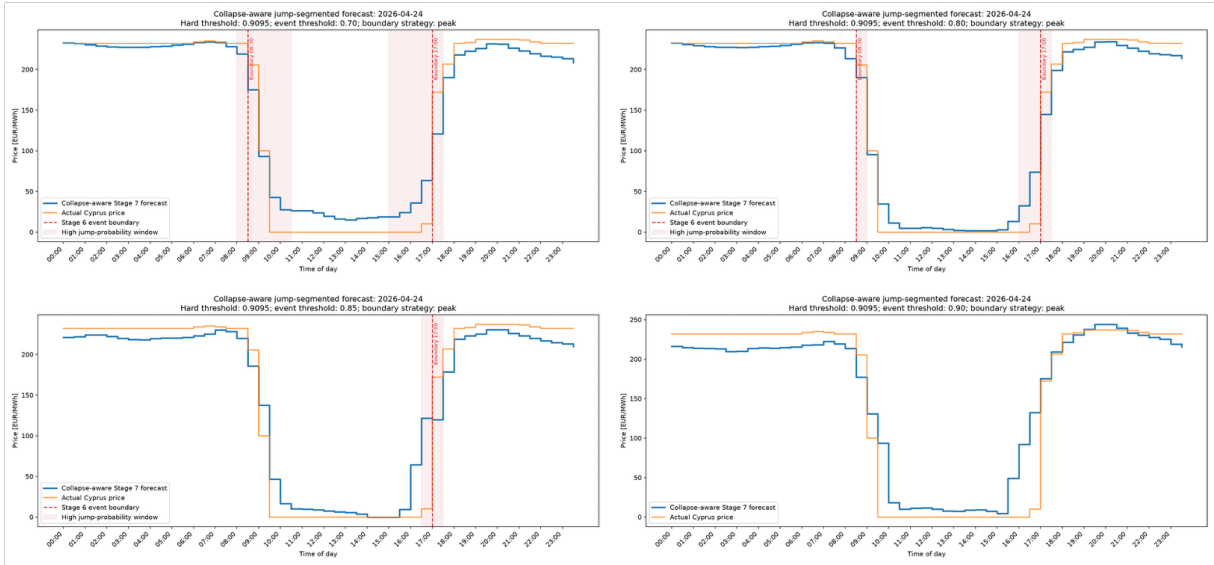


Figure 7: Example of the collapse-aware peak-boundary strategy on 24 April 2026. Each high-probability event window contributes at most one boundary, placed at the half-hour with the largest Stage 6 jump probability inside that window.

The examples above were selected because they clearly illustrate the behaviour of each approach and are not intended to form a direct visual comparison with one another. Matched start-stop and peak plots for the same target dates are provided in Appendix B.

Historical Analogue Blending and Clipping

Although the historical data contained a common daily pattern of midday price decline and subsequent recovery, the timing, depth, and speed of this movement varied between days. Particularly sharp or deep collapses occurred less frequently. A transformer trained across many days could therefore predict a more moderate version of the usual pattern and underestimate an unusually large price change.

To address this problem, the collapse-aware method added a historical analogue forecast to high-risk segments. This forecast was constructed using earlier days with market conditions similar to those of the target day. A segment was considered high-risk when its maximum Stage 6 jump probability was at least 0.80.

The target segment’s standardised forecast variables were compared with the corresponding clock segment on every available historical day using mean squared feature distance. The 20 most similar historical days were selected. For each half-hour in the segment, the median price at that same time across the 20 selected days was used to construct the historical analogue forecast.

The final forecast for a high-risk segment was calculated as a weighted combination of 65% of the transformer forecast and 35% of the historical analogue forecast, allowing the transformer to remain the main forecast component while incorporating information from historically similar market conditions.

For Cyprus, the collapse-aware implementation clipped every segment prediction to the minimum and maximum prices observed in the permitted pre-target history. This prevented the model from producing unintended negative or otherwise out-of-range forecasts. However, it also meant that the model could not predict a genuinely new historical extreme. The initial hard-boundary approach did not apply this price clipping.

Collapse-Aware Threshold Comparison

The start-stop event threshold was evaluated across all 61 target days from April and May 2026.

Event threshold	Mean MAE [€/MWh]	Median MAE [€/MWh]	Mean RMSE [€/MWh]	Median RMSE [€/MWh]	Avg event boundaries/day	Zero-event days	Avg segments/day
0.70	17.501	16.102	29.536	27.323	2.721	0	3.721
0.80	16.849	15.812	29.708	30.006	2.082	0	3.082
0.85	19.512	17.108	31.781	29.328	1.656	8	2.656
0.90	20.487	17.431	31.961	28.206	1.115	19	2.115

Table 4: Collapse-aware start-stop event-threshold sensitivity for April–May 2026.

The 0.80 threshold produced the lowest mean MAE. Threshold 0.70 produced a slightly lower mean RMSE but also created more boundaries and required more segment models. Threshold 0.80 was therefore selected as the practical default because it achieved the lowest mean MAE with a lower computational cost and a similar mean RMSE.

Performance deteriorated at thresholds 0.85 and 0.90. These settings produced fewer boundaries and resulted in eight and nineteen days, respectively, with no accepted event boundary. This suggests that stricter event thresholds discarded some probability regions that were potentially useful to the price forecaster.

Threshold 0.80 was selected after comparing the results of all four thresholds on the April–May dates. It should therefore be described as the best practical choice for this particular period, rather than as the best threshold in general. A stronger future evaluation would select the threshold using an earlier validation period and then test the chosen value on a later period that was not involved in the selection.

The peak strategy, which considered one maximum-probability boundary per event window, was also evaluated across the April–May dates.

Event threshold	Mean MAE [€/MWh]	Median MAE [€/MWh]	Mean RMSE [€/MWh]	Median RMSE [€/MWh]	Avg event boundaries/day	Zero-event days	Avg segments/day
0.70	19.391	16.503	31.150	29.380	1.475	0	2.475
0.80	19.785	15.667	31.133	29.589	1.557	0	2.557
0.85	19.826	16.091	31.127	28.734	1.230	8	2.230
0.90	20.102	15.722	31.800	29.244	0.836	19	1.836

Table 5: Collapse-aware peak-boundary event-threshold sensitivity for April–May 2026.

The best peak mean MAE was obtained at threshold 0.70, while its best mean RMSE was obtained at threshold 0.85. Both values were worse than the best start-stop results.

A direct comparison of the two boundary strategies is shown below.

Event threshold	Start-stop mean MAE [€/MWh]	Peak mean MAE [€/MWh]	Start-stop mean RMSE [€/MWh]	Peak mean RMSE [€/MWh]
0.70	17.501	19.391	29.536	31.150
0.80	16.849	19.785	29.708	31.133
0.85	19.512	19.826	31.781	31.127
0.90	20.487	20.102	31.961	31.800

Table 6: Comparison of start-stop and peak-probability boundary approaches for April–May 2026.

The results suggest that, for this dataset, it was more useful to retain the full high-probability event window rather than reducing it to one maximum-probability half-hour. The start-stop strategy allowed the price model to treat the interval containing the decline and recovery as a separate segment. The peak strategy divided the day once inside that interval and did not explicitly mark both sides of it.

Start-stop was not better at every individual matched threshold, particularly at the highest threshold. Nevertheless, it produced the best overall mean MAE and RMSE and was therefore retained as the recommended boundary strategy.

Effect of Stage 6 Errors on Stage 7

Stage 7 depended on the accuracy and timing of the Stage 6 jump probabilities. A false high-probability event could create an unnecessary boundary, while failing to assign a high probability to a genuine Jump could cause different price periods to be forecasted as a single segment. Stage 6 could also identify the general period of a price transition without identifying its exact time. In this case, the selected boundary could occur earlier or later than the actual price transition. The collapse-aware method reduced the oversegmentation by grouping neighbouring high-probability half-hours, but it did not remove this dependence on Stage 6.

Results and Interpretation

The daily forecast plots showed promising results on some dates, particularly for the start-stop strategy, which captured the general timing and shape of several price declines and recoveries.

The initial ten-day hard-boundary comparison produced its lowest mean MAE and mean RMSE at threshold 0.80, with values 25.556 €/MWh and 39.530 €/MWh, respectively. Across the 61-day collapse-aware experiment, start-stop obtained its lowest mean MAE of 16.849 €/MWh at event threshold 0.80 and its lowest mean RMSE of 29.536 €/MWh at event threshold 0.70.

These results should not be treated as a controlled before-and-after comparison. The hard-boundary experiment covered only ten days, whereas the collapse-aware experiment covered 61 days and also added historical analogue blending and historical-range clipping. The results show that the collapse-aware setting performed well in the saved experiments, but they do not measure the isolated effect of event clustering alone.

Reproducibility of the Strategy Comparisons

The forecasting script set the random seed to 42 once at the beginning of each complete experiment. The random seed controlled the model’s initial weights and the order of the training examples. However, the script did not reset the seed before training each date and segment model.

Consequently, two experiments could produce different forecasts for a later date even when they selected the same boundaries on that date. This could happen because the experiments may have created different numbers or lengths of segments on earlier dates and therefore trained different numbers of models. Although both experiments started with the same random seed, these earlier differences meant that the models trained later could receive different starting weights and process their training examples in different orders.

For example, the start-stop and peak experiments could both produce one full-day segment on a particular date. However, if they had produced different segments on earlier dates, the models trained for the full-day segment would not necessarily start with the same weights or process the training examples in the same order. Their forecasts could therefore differ even though they used identical boundaries on the date being examined.

This was a training reproducibility effect rather than an intended effect of the boundary strategy. A more controlled comparison would reset the seed before training each segment, using a value based only on the target date and segment boundaries. Identical date-and-segment forecasting tasks would then receive the same model initialisation and order of training examples.

Appendix B provides matched start-stop and peak plots for the same target dates. These plots should be interpreted with this random-seed limitation in mind.

Application to the German Market

An additional experiment applied the collapse-aware start-stop method to the German market. It used the German Stage 6 rolling probabilities and divided each day into 24 hourly periods rather than 48 half-hourly periods.

The German experiment used an event threshold of 0.70 and did not apply price clipping. This was necessary because negative electricity prices are legitimate observations in the German market. Applying the Cyprus clipping assumptions would therefore have been inappropriate.

The German experiment showed that the Stage 7 procedure could be applied to another electricity market. However, it also showed that the event threshold, segmentation settings, and clipping rules should be chosen separately for each market.

Limitations and Future Work

The main limitations of Stage 7 were:

- A separate transformer was trained for every segment and target date, resulting in substantial runtime.
- The fixed boundary-spacing and maximum-boundary rules could discard short or closely spaced price events, while inaccurate Stage 6 jump probabilities could place boundaries earlier or later than the actual price transitions.
- For October 2025 to March 2026, the Stage 6 jump probability feature was generated using the model trained for the final Stage 6 target date. These historical probabilities therefore did not follow a strictly chronological procedure.
- The segment models did not use a validation set or early stopping, and the model settings such as the 14-day lookback were not systematically optimised.
- The metric used to measure similarity between the target segment and historical segments, the number of most similar historical segments selected, the analogue blend weight, and the historical-range clipping rules were fixed manually.
- The random seed was not reset separately for each date and segment, which weakened direct comparison between experiments. Matched segment models in different runs did not always receive identical initialisation and training-example ordering.
- The main Cyprus evaluation covered only 61 days, limiting the conclusions that could be drawn about performance in other seasons and market conditions.

Future work should improve the chronological construction of the Stage 6 jump probability feature. Each historical date’s probabilities should be generated using a classifier trained only on information available before that date. This would ensure that both the Stage 6 and Stage 7 parts of the complete pipeline follow the same chronological evaluation procedure.

The event threshold and other model settings should be selected using an earlier validation period and then evaluated on a later period. This evaluation could compare different lookback lengths, minimum segment lengths, maximum boundary counts, transformer settings, analogue-selection rules, blend weights, and clipping rules.

Resetting a date-and-segment specific random seed before training every segment model would make comparisons between thresholds and boundary strategies more controlled. Runtime could be reduced by training one shared model across dates or segments rather than fitting a new transformer for every segment.

Additional market information, such as energy-forward contract quantities, could be included as model inputs. Finally, the model should be evaluated using larger electricity-market datasets.

The German experiment provided an initial transferability test, but further experiments would be needed to determine how reliably the approach generalises across different market structures, time resolutions, and price behaviours.

Conclusion

This research internship investigated the use of transformer-based models to analyse electricity-price behaviour and forecast sharp movements in Day-Ahead electricity prices. The seven-stage workflow progressed from controlled experiments on synthetic time series to the use of predicted jump scores to guide electricity-price forecasting using real market data.

Stage 6 examined whether electricity-market forecast variables could identify intervals associated with subsequent sharp price changes. The results indicated that these variables may contain information relevant to prediction, but also revealed a substantial precision-recall trade-off.

Stage 7 incorporated the Stage 6 jump scores into a segmented price-forecasting framework. Across the 61 evaluated target days, the start-stop strategy achieved its lowest mean MAE at an event threshold of 0.80 and its lowest mean RMSE at a threshold of 0.70. It performed better overall than the peak-boundary strategy. However, these results should not be interpreted as a controlled demonstration that the collapse-aware method outperformed the initial implementation, because the two implementations were evaluated under different experimental conditions.

Overall, the project established a complete exploratory pipeline connecting electricity-market forecasts to sharp-movement classification and segmented price forecasting. Although further validation is required before firm conclusions can be drawn, the work provides a foundation for investigating whether anticipated market transitions can support Day-Ahead electricity-price forecasting.

Additional Notes

In addition to the principal workflows presented in this report, the project included supporting utilities, preliminary extensions, and exploratory comparisons across several stages. Some of these are noted below to clarify their purpose and status, while further details are provided in the documentation accompanying the source code.

Stage 6 includes a future-inference utility that applies a saved row-wise classifier to new market forecast data. By default, it uses the most recent saved checkpoint and produces jump probabilities and predicted labels without requiring realised prices. This is appropriate for dates after the checkpoint’s training period. It should not be used retrospectively for earlier dates because the checkpoint may have been trained using observations that occurred after the date being predicted. Constructing chronologically valid probabilities for earlier historical dates would require a separate rolling classifier trained for each date using only information available before that date.

The German Stage 6 and Stage 7 workflows were retained as transferability experiments. Detailed German results and interpretation are provided with the corresponding project outputs.

They are not presented extensively in this report because Cyprus was the principal case study and received the main evaluation.

Stage 7 contains a preliminary extension that adds historical forward-contract energy quantity to the collapse-aware forecasting inputs. The available forward-quantity data began in January 2026, whereas the principal Stage 7 workflow used history beginning in October 2025. The extension therefore used a shorter historical period, and saved results were retained only for 1–4 April 2026. This implementation demonstrates how the feature could be incorporated, but the limited run does not establish whether it improves forecasting. A complete comparison would require a longer consistent dataset and matched evaluation conditions.

The original Stage 7 implementation also supports a historical-analogue baseline method that does not train or use a transformer. This method selects past segments with similar market forecast conditions and uses their median price profile as the prediction. This is separate from the historical-analogue blend used by the principal collapse-aware model and was retained as an optional comparison rather than a main result.

Further Stage 7 architectural limitations documented in the project include:

- each historical day is compressed into one transformer token, so attention compares days rather than individual half-hourly observations;
- mean pooling gives every target position in a segment the same overall summary of the historical input;
- the principal model does not include explicit calendar variables; and
- the scripts are structured primarily for historical evaluation rather than as a fully separated future-production interface.

These secondary implementations and limitations provide directions for future development but do not change the interpretation of the principal Cyprus results.

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A Stage 6 Confusion-Count Curves

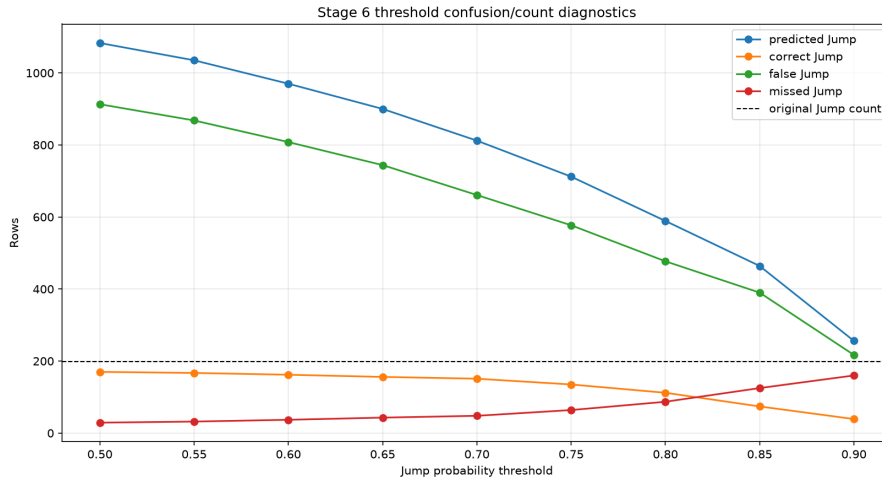


Figure 8: Stage 6 prediction and confusion counts across different jump-probability thresholds for the Cyprus April–May 2026 evaluation period. The curves show the number of predicted Jumps, correctly detected Jumps, false Jumps, and missed Jumps. The dashed horizontal line represents the 199 original Jump intervals. As the threshold increased, the number of predicted and false Jumps decreased, but fewer original Jumps were correctly detected and more were missed.

B Matched Stage 7 Daily Forecasts

This appendix presents forecasts produced by different Stage 7 segmentation strategies for the same target dates. Comparing methods on matched dates helps illustrate how the selected boundaries affect the final price forecast. Because the random seed was not reset before every segment-level training run, forecasts may still differ slightly even when two strategies produce identical boundaries. The accompanying project files contain forecast plots for all evaluated dates: the first ten days of April 2026 for the initial approach and all 61 days from April to May 2026 for both collapse-aware strategies.

1 April 2026

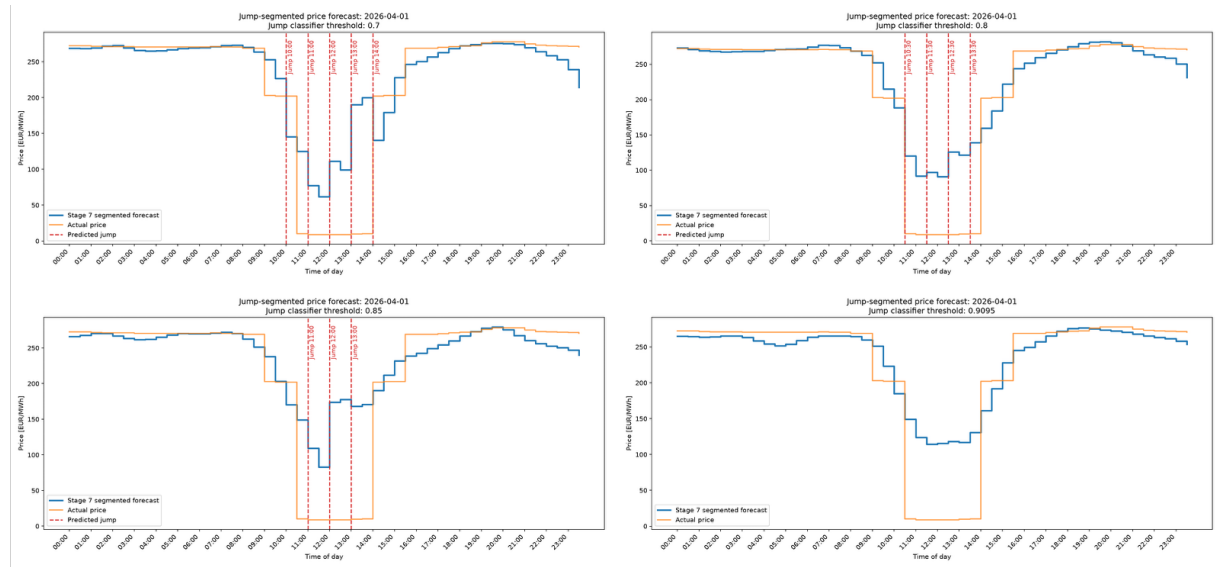


Figure 9: Initial hard-boundary Stage 7 forecast for 1 April 2026.

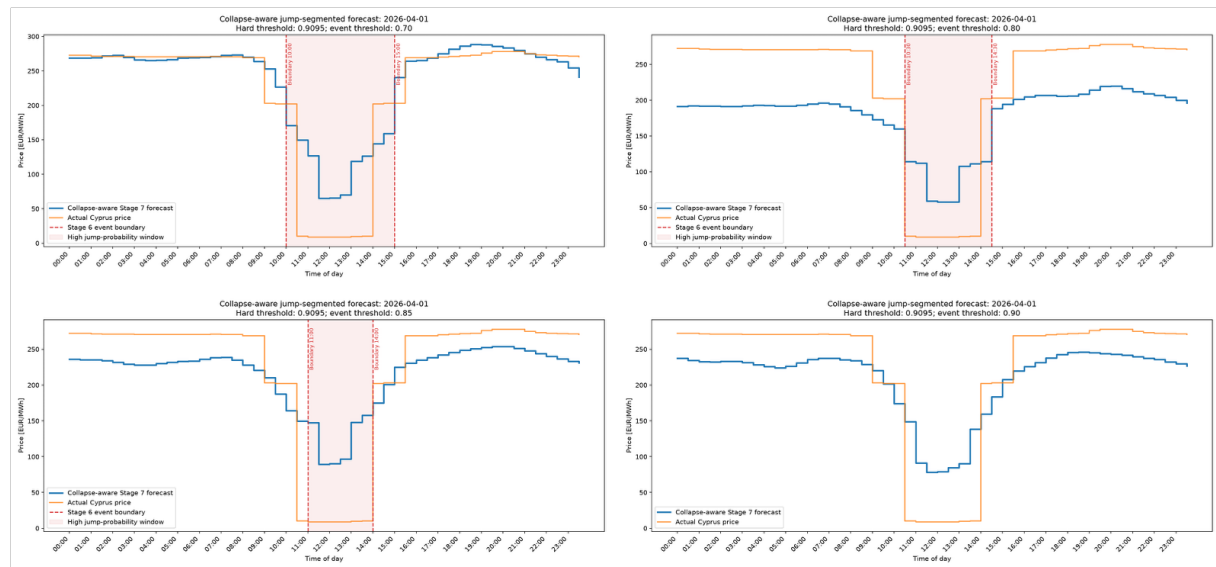


Figure 10: Collapse-aware start-stop Stage 7 forecast for 1 April 2026.

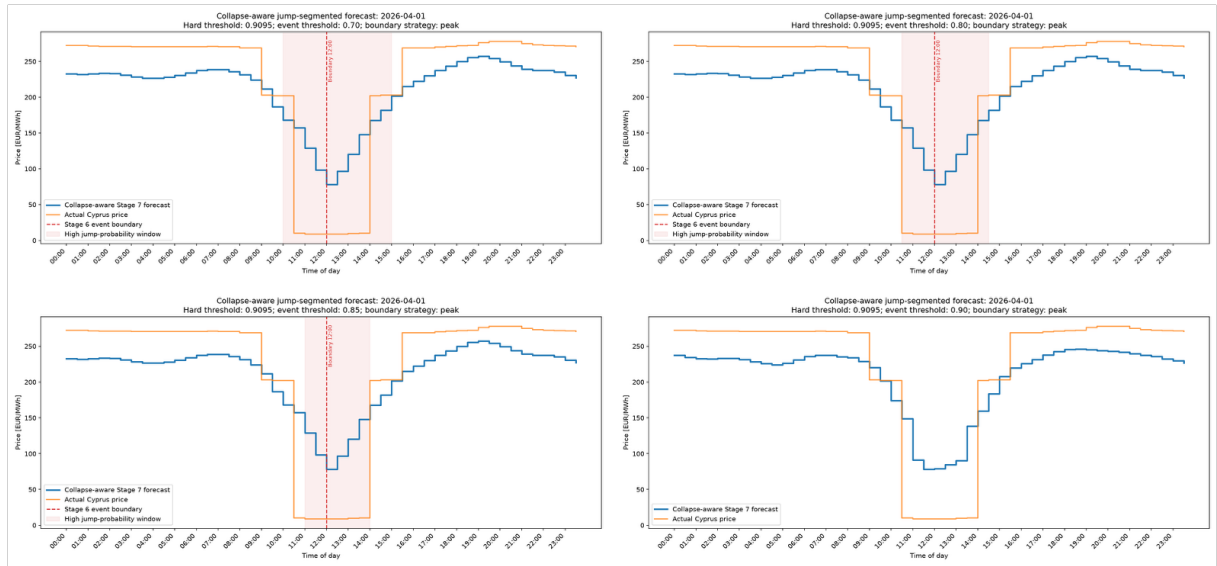


Figure 11: Collapse-aware peak-boundary Stage 7 forecast for 1 April 2026.

15 May 2026

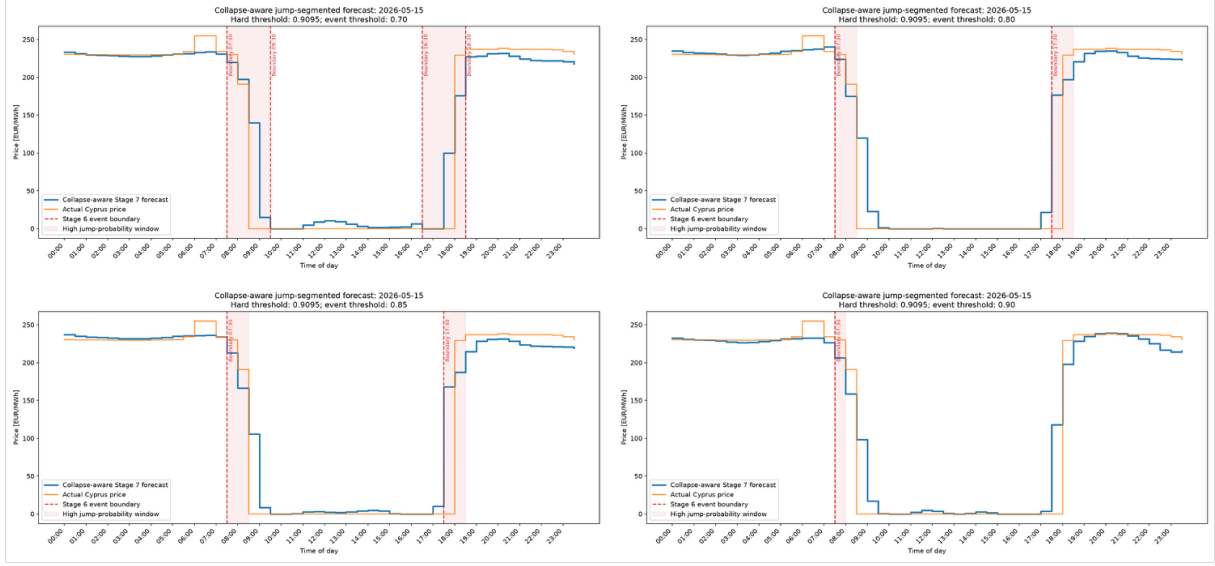


Figure 12: Collapse-aware start-stop Stage 7 forecast for 15 May 2026.

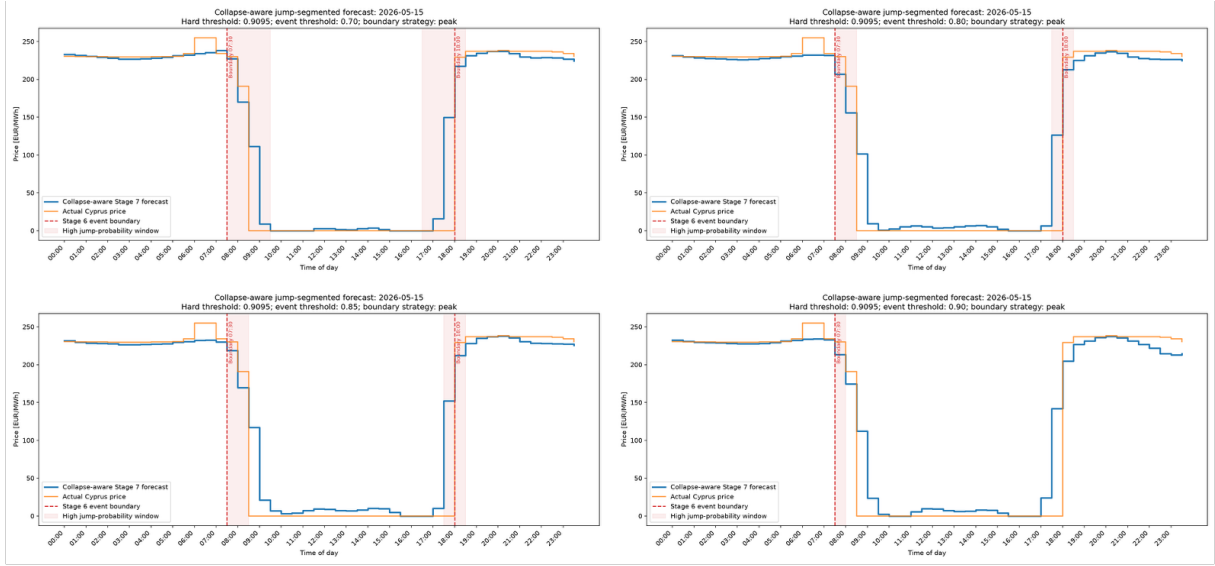


Figure 13: Collapse-aware peak-boundary Stage 7 forecast for 15 May 2026.

24 April 2026

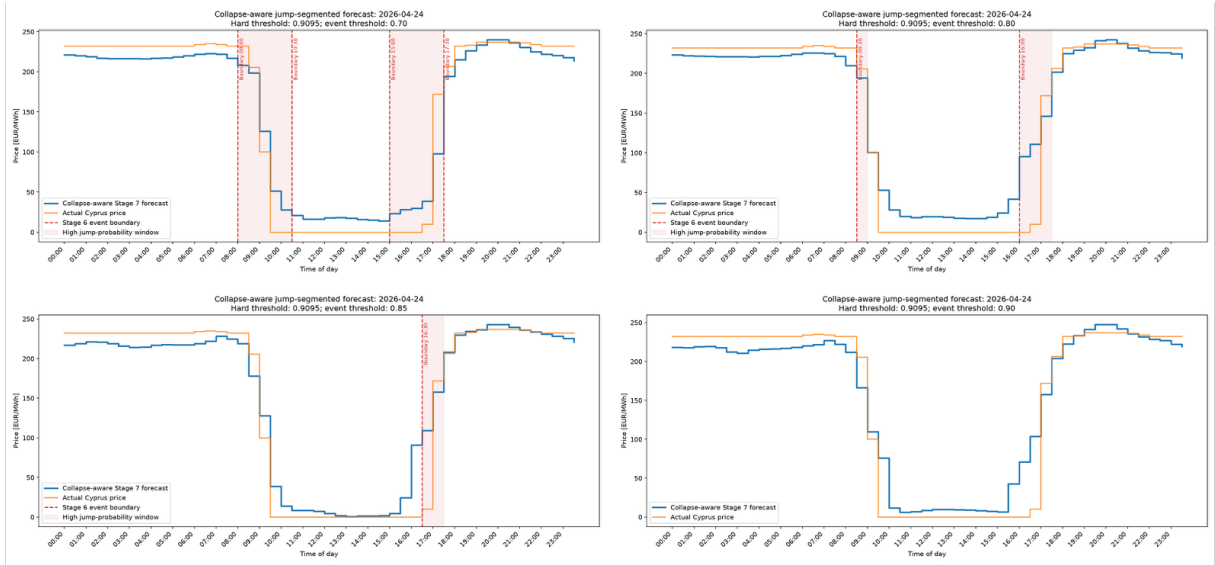


Figure 14: Collapse-aware start-stop Stage 7 forecast for 24 April 2026.

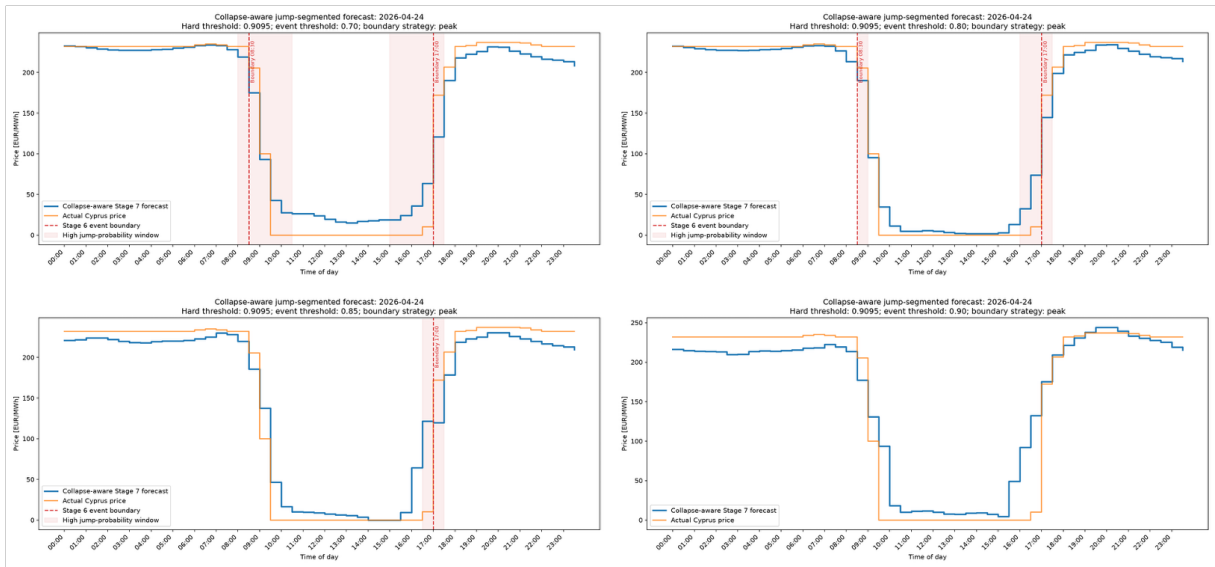


Figure 15: Collapse-aware peak-boundary Stage 7 forecast for 24 April 2026.

C Software and Project Materials

The project was implemented in Python. The principal packages were PyTorch for model development, NumPy and pandas for numerical and time-series data processing, Matplotlib for visualisation, and openpyxl for spreadsheet input and output.

The complete project repository will be supplied as part of the handover, including the source code for all seven stages, data-processing and experimental scripts, model outputs, evaluation tables, generated plots, and documentation.